

Which pairs of substances are released from exocrine glands?

- ☐ A. Glucagon and tears [4%]
- ☐ B. Bile and insulin [6%]
- ☐ C. Insulin and digestive enzymes [11%]
- ✓ ☒ D. Tears and bile [76%]

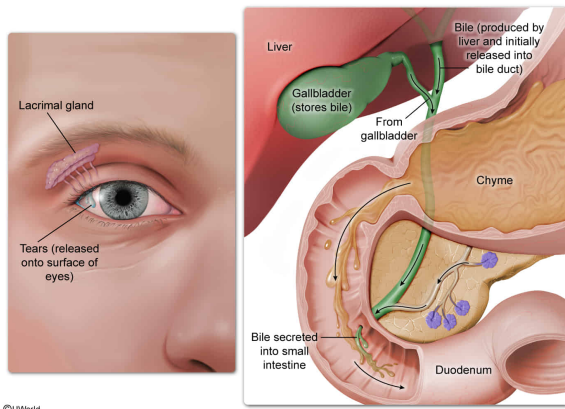
Correct

76% answered correctly

Explanation:

Tears and bile are exocrine secretions

Exocrine secretions are released onto an "exterior" body surface (eg, eye surface, lumen on duodenum)



Certain types of **epithelial cells** are specialized for the secretion of substances into the extracellular space that surrounds them. Some of these secretory epithelial cells are solitary, being one of many epithelial cell types distributed across the surface of a tissue, whereas others are grouped together in a collection of secreting cells called a **gland**.

During development, some secretory cells form hollow tubes called **ducts** into which their secretions flow. For example, mammary glands secrete breast milk through ducts connected to the body's exterior via the nipple, and **sweat glands** secrete sweat into ducts that empty onto the skin's surface via pores. Glands whose secretions are released onto an "exterior" body surface (such as the skin or the inner surface of the lung) are called **exocrine glands**.

Other secretory epithelial cells develop around blood vessels but lack ducts; these cells release their secretions, called **hormones**, into the blood. Glands composed of cells such as these are called **endocrine glands**.

Some organs contain both endocrine and exocrine cells. For example, the **pancreas** secretes both digestive enzymes (an exocrine secretion) and **insulin and glucagon**, which are hormones (endocrine secretions).

The question asks for the pair of substances that would be secreted from exocrine glands. **Tears** are **secreted onto** the **eye's surface** from lacrimal glands near the eyes,

and **bile** is **secreted from** the **liver into** the **bile duct**, from which it is **released into** the **duodenum** of the small intestine (recall that the epithelial surfaces of the stomach, intestines, and lungs border cavities that are in communication with the exterior of the body). Therefore, **tears and bile** are both **secreted from exocrine glands**.

(Choices A, B, and C) Insulin and glucagon are hormones. Because hormones are secreted into the blood from endocrine glands, these choices are not examples of pairs of substances secreted from exocrine glands.

Educational objective:

Both endocrine and exocrine cells are epithelial cells that produce substances released into the extracellular space. Exocrine secretions are released to the body's "exterior," such as the skin or the inner surface of the gut or lungs. Endocrine secretions are released into the blood.

Time spent:0 sec

Subject:
Biology

QID:404158

System:
Endocrine and Nervous Systems

Which table shows the typical neurotransmitters released by preganglionic and postganglionic neurons of sympathetic and parasympathetic pathways?

☐ A.

Pathway	Site of release	Typical neurotransmitter
Sympathetic	Preganglionic	Norepinephrine
	Postganglionic	Acetylcholine
Parasympathetic	Preganglionic	Norepinephrine
	Postganglionic	Acetylcholine

[12%]

☐ B.

Pathway	Site of release	Typical neurotransmitter
Sympathetic	Preganglionic	Acetylcholine
	Postganglionic	Norepinephrine
Parasympathetic	Preganglionic	Acetylcholine
	Postganglionic	Norepinephrine

[10%]

☐ C.

Pathway	Site of release	Typical neurotransmitter
Sympathetic	Preganglionic	Acetylcholine
	Postganglionic	Acetylcholine
Parasympathetic	Preganglionic	Acetylcholine
	Postganglionic	Norepinephrine

[11%]

☒ D.

Pathway	Site of release	Typical neurotransmitter
Sympathetic	Preganglionic	Acetylcholine
	Postganglionic	Norepinephrine
Parasympathetic	Preganglionic	Acetylcholine
	Postganglionic	Acetylcholine

[65%]

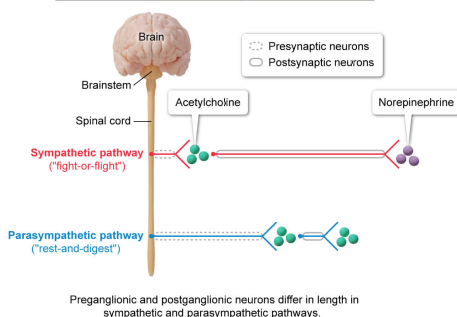
Correct

64% answered correctly

Explanation:

Postganglionic neurotransmitters typically differ in sympathetic and parasympathetic pathways

Pathway	Site	Typical neurotransmitter
Sympathetic	Preganglionic	Acetylcholine
	Postganglionic	Norepinephrine
Parasympathetic	Preganglionic	Acetylcholine
	Postganglionic	Acetylcholine



Within the **peripheral nervous system**, **afferent nerves** carry sensory information from the periphery

back to the central nervous system, and **efferent nerves** **transmit neural impulses** from the central nervous system to target tissues. The efferent nerves make up the **efferent (motor) division** of the peripheral nervous system. The **autonomic nervous system**, one of the two subdivisions of the motor division, can itself be further divided into **two branches**:

- The **sympathetic nervous system** mediates involuntary "fight-or-flight" responses, such as

increasing heart rate and muscle blood flow in response to a startling stimulus.

- The **parasympathetic nervous system** mediates involuntary "rest-and-digest" responses during nonstressful situations, such as promoting a slower heart rate while at rest.

A **ganglion** is a collection of several nerve cell bodies within the peripheral nervous system and is the site where preganglionic nerves synapse with postganglionic nerves. Both sympathetic and parasympathetic responses are transmitted along preganglionic fibers arising from the central nervous system and postganglionic fibers that synapse with the target tissue(s) responsible for a particular response.

The question asks for the table showing the typical neurotransmitters released by preganglionic and postganglionic neurons of sympathetic and parasympathetic pathways. **Both sympathetic and parasympathetic preganglionic neurons release acetylcholine** from their nerve terminals. However, the **neurotransmitter released at the target tissue** by the postsynaptic neuron **varies** for sympathetic and parasympathetic pathways, with **acetylcholine** typically **released in parasympathetic** pathways and **norepinephrine** typically **released in sympathetic** pathways.

(Choices A, B, and C) The tables in these choices indicate *norepinephrine* released from preganglionic neurons, norepinephrine released from *parasympathetic* postsynaptic neurons, or acetylcholine released from *sympathetic* postsynaptic neurons, all of which are incorrect.

Educational objective:

Sympathetic pathways mediate "fight-or-flight" responses whereas parasympathetic pathways mediate "rest-and-digest" pathways. Parasympathetic pathways typically involve acetylcholine released from both pre- and postganglionic neurons. Sympathetic pathways also employ acetylcholine released from preganglionic neurons but typically release norepinephrine from postsynaptic neurons.

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Biology

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System:
Endocrine and Nervous Systems

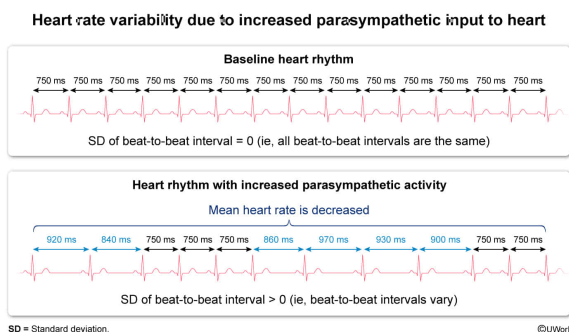
The interval between successive heart beats is modulated by the balance between sympathetic and parasympathetic input to the heart. Given this, an increase in parasympathetic input that increases variability in heart rate will most likely have which of the following effects?

- ☐ A. The mean heart rate and the standard deviation of the interval between beats will both increase. [8%]
- ☐ B. The mean heart rate and the standard deviation of the interval between beats will both decrease. [14%]
- ✓ ☒ C. The mean heart rate will decrease and the standard deviation of the interval between beats will increase. [72%]
- ☐ D. The mean heart rate will increase and the standard deviation of the interval between beats will decrease. [4%]

Correct

71% answered correctly

Explanation:



The **autonomic** nervous system **transmits neural impulses** from the central nervous system to target tissues throughout the body to regulate involuntary processes (eg, **digestion**). The relative activities of the **two branches** of the autonomic nervous system (**sympathetic** and **parasympathetic**) determines the balance between the sympathetically mediated "fight-or-flight" responses (eg, more forceful contractions by the heart) and the parasympathetically mediated "rest-and-digest" responses (eg, pupil constriction).

The values measured in an experiment (eg, the time interval between heart beats) can be summarized in various ways. A statistic summarizing the central tendency of the values (eg, **mean**, **median**, **mode**) is often calculated, along with a statistic indicating the spread of the values. For example, the **standard deviation** is a statistic indicating how far the values are spread around the mean.

The question first states that the balance between sympathetic and parasympathetic input to the heart modulates heart rate variability, and then asks for the likely effects of an increase in parasympathetic input that increases variability in heart rate.

Parasympathetic stimulation of the **heart decreases** the **heart rate**, and **greater**

variability in heart rate means that the interval between successive heart beats is more variable. Therefore, the **standard deviation**, a measure of the spread of individual values, **will increase**.

(Choices A and D) Increased parasympathetic activity will decrease (*not* increase) the heart rate.

(Choice B) Increased variability will increase (*not* decrease) the standard deviation.

Educational objective:

The parasympathetic nervous system is one of the branches of the autonomic nervous system and mediates "rest-and-digest" responses. Parasympathetic stimulation of the heart reduces the heart rate. The standard deviation is a measure of the spread of values around their mean.

Time spent:0 sec

Subject:
Biology

QID:404160

System:
Endocrine and Nervous Systems